

For this role, the technology stack spans **AI/LLM engineering, cloud engineering, backend development, distributed systems, DevOps, and enterprise architecture.** Below is a comprehensive breakdown of the technical skills required.

## 1. AI & Generative AI (Highest Priority)

### LLM Fundamentals

- Transformer architecture
- Attention mechanism
- Tokenization
- Context window
- Embeddings
- Fine-tuning vs Prompt Engineering
- Model selection
- Function Calling
- Structured Outputs
- Streaming responses

### Models

- GPT-4.x
  - GPT-5 family
  - Claude
  - Gemini
  - Llama
  - Mistral
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## 2. Azure AI Foundry

Need strong hands-on knowledge of:

- Agent Service
- Prompt Flow
- Model Catalog
- Azure OpenAI
- Evaluation
- AI Search integration
- Fine-tuning
- Content Safety
- Monitoring
- Deployment
- SDK

- REST APIs
- Managed Identity

Should understand

- AI Projects
  - AI Hub
  - Connections
  - Prompt Templates
  - Evaluators
  - Tracing
  - Telemetry
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### 3. AI Agent Frameworks

Must know at least two well.

#### **Semantic Kernel**

- Plugins
- Skills
- Memory
- Planner
- Function Calling
- Agent orchestration

#### **LangChain**

- Chains
- Agents
- Tools
- Memory
- LCEL
- LangGraph
- Retrieval
- Output Parsers

#### **AutoGen**

- Multi-agent conversations
- Task planning
- Tool usage
- Group Chat
- Human-in-the-loop

Also useful

- CrewAI
  - Microsoft Agent Framework
  - OpenAI Agent SDK
  - LlamaIndex
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## 4. Multi-Agent Systems

Understand

- Planner Agent
- Executor Agent
- Critic Agent
- Reviewer Agent
- Supervisor Agent
- Routing Agent

Patterns

- Hierarchical agents
  - Swarm
  - Sequential
  - Parallel
  - Event-driven
  - Reflection
  - ReAct
  - Tree of Thoughts
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## 5. RAG (Retrieval Augmented Generation)

Need deep knowledge.

Topics

- Embeddings
- Chunking
- Metadata
- Hybrid Search
- Semantic Search
- Keyword Search
- Reranking

- Vector Database
- Grounding
- Context Injection

#### Vector databases

- Azure AI Search
  - Pinecone
  - Qdrant
  - Chroma
  - FAISS
  - Milvus
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## 6. Prompt Engineering

#### Need to know

- System prompts
  - User prompts
  - Chain of Thought
  - Few-shot prompting
  - Zero-shot prompting
  - Role prompting
  - XML prompting
  - JSON outputs
  - Prompt optimization
  - Prompt evaluation
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## 7. Backend Development

Python preferred.

#### Should know

- FastAPI
- Flask
- Async Programming
- REST APIs
- WebSockets
- Streaming APIs
- Dependency Injection

If Java

- Spring Boot
  - Spring AI
  - REST APIs
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## 8. API Integration

Must understand

- REST
- GraphQL
- OAuth
- JWT
- API Gateway
- Webhooks

Integration

- Microsoft Graph
  - SharePoint
  - SAP
  - Salesforce
  - ServiceNow
  - Databases
  - CRM
  - ERP
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## 9. Cloud (Azure)

Need strong Azure experience.

Services

- Azure Functions
- Azure Storage
- Azure App Service
- Azure Container Apps
- AKS
- Azure Kubernetes Service
- Azure AI Search
- Azure Service Bus

- Event Grid
  - Event Hub
  - Cosmos DB
  - Azure SQL
  - Key Vault
  - Azure Monitor
  - Application Insights
  - Azure DevOps
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## 10. Microservices

Need

- REST APIs
  - Event-driven architecture
  - Domain-driven design
  - API Gateway
  - CQRS
  - Saga Pattern
  - Circuit Breaker
  - Retry
  - Message Queue
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## 11. Event-Driven Systems

Should know

- Azure Service Bus
- RabbitMQ
- Kafka
- Azure Event Grid

Concepts

- Pub/Sub
  - Queue
  - Topic
  - Dead Letter Queue
  - Retry
  - Event Sourcing
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## 12. Databases

### SQL

- SQL Server
- PostgreSQL
- MySQL

### NoSQL

- Cosmos DB
- MongoDB

### Caching

- Redis

### Vector DB

- Azure AI Search
  - Pinecone
  - Qdrant
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## 13. DevOps

### Need

- Azure DevOps
  - GitHub Actions
  - CI/CD
  - YAML pipelines
  - Release pipelines
  - Testing
  - Deployment automation
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## 14. Containers

### Need

- Docker
  - Docker Compose
  - Kubernetes
  - AKS
  - Helm
  - Container Registry
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## 15. Security

Need understanding of

- OAuth
- OpenID Connect
- JWT
- RBAC
- Managed Identity
- Azure Key Vault
- Encryption
- Secrets Management

AI Security

- Prompt Injection
  - Jailbreak protection
  - Data Leakage
  - Content Filtering
  - Responsible AI
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## 16. Monitoring

Need

- Azure Monitor
  - Application Insights
  - OpenTelemetry
  - Logging
  - Metrics
  - Tracing
  - AI Evaluation
  - Cost Monitoring
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## 17. Performance Optimization

Need

- Token optimization
- Caching
- Async execution
- Batch processing
- Parallel execution



- Streaming
  - Model routing
  - Latency optimization
  - Cost optimization
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## 18. Testing

Need

- PyTest
  - Unit Testing
  - Integration Testing
  - Mocking
  - AI Evaluation
  - Prompt Testing
  - Regression Testing
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## 19. Frontend (Nice to Have)

- React
  - TypeScript
  - Chat UI
  - Streaming UI
  - Markdown rendering
  - Authentication
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## 20. Architecture Skills

Should know

- Solution Architecture
  - Enterprise Architecture
  - Cloud Architecture
  - AI Architecture
  - Distributed Systems
  - Scalability
  - High Availability
  - Fault Tolerance
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## 21. Software Engineering Best Practices

### Need

- SOLID Principles
- Design Patterns
- Clean Architecture
- Hexagonal Architecture
- Repository Pattern
- Dependency Injection
- Async Programming
- Git

## 22. Enterprise AI Governance

### Need

- Responsible AI
- Explainability
- Compliance
- Monitoring
- Model Evaluation
- Bias Detection
- Auditability

## Ideal Technology Stack for This Role

| Category         | Technologies                                          |
|------------------|-------------------------------------------------------|
| Language         | Python, Java                                          |
| AI Platform      | Azure AI Foundry, Azure OpenAI                        |
| Agent Frameworks | Semantic Kernel, LangChain, AutoGen, OpenAI Agent SDK |
| LLM              | GPT, Claude, Gemini, Llama                            |
| RAG              | Azure AI Search, FAISS, Pinecone, Qdrant              |
| Backend          | FastAPI, Flask, Spring Boot                           |
| Cloud            | Azure Functions, AKS, App Service, Storage, Cosmos DB |

| Category   | Technologies                                       |
|------------|----------------------------------------------------|
| Messaging  | Azure Service Bus, Kafka, RabbitMQ                 |
| DevOps     | Azure DevOps, GitHub Actions                       |
| Containers | Docker, Kubernetes                                 |
| Monitoring | Application Insights, Azure Monitor, OpenTelemetry |
| Security   | OAuth, Managed Identity, Key Vault                 |
| Frontend   | React, TypeScript                                  |

## Most Critical Technologies (Priority Order)

If you have limited time, focus on these in order:

1. Python
2. Azure AI Foundry
3. Azure OpenAI
4. Semantic Kernel
5. LangChain/LangGraph
6. Retrieval-Augmented Generation (RAG)
7. FastAPI
8. Azure AI Search
9. Azure Functions
10. Docker
11. Kubernetes (AKS)
12. Azure Service Bus
13. Azure DevOps (CI/CD)
14. OpenTelemetry & Application Insights
15. Multi-agent architectures and orchestration

Mastering these technologies will cover the majority of the technical expectations for a senior Software Engineer working on enterprise Agentic AI solutions with Azure.